SELF ASSESSMENT 2 - 3 (Page 56)

1. Process $1-2$ $S_1 = S_2$; $Q_{12} = 0$

$$\gamma_{He} = 1.667,\ R_{He} = 2.0771 \text{kJ/kg K},\ C_V = R_{He}/(\gamma - 1),\ 0^0\text{C} = 273\ \text{K}$$

$Q_{12} + W_{12} = mC_v(T_2 - T_1)$ since $\Delta KE = \Delta PE = 0$

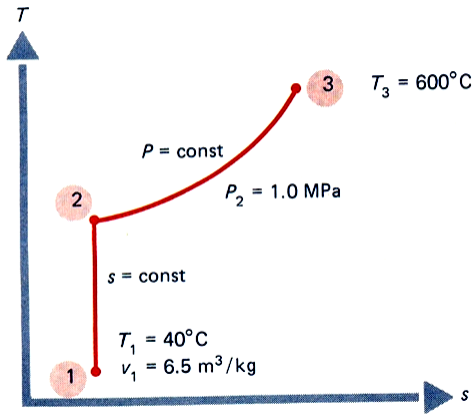
Process $1-2$ is $PV^\gamma = \text{constant}$

$P_1V_1^\gamma = P_2V_2^\gamma$

$$\left(\frac{V_2}{V_1}\right)^\gamma = \left(\frac{P_1}{P_2}\right) = \frac{100.07}{1000}$$

$$W_{12} = \frac{1 \times 2.0771}{(1.667 - 1)} \times [786.60 - 313.15]$$

$|W_{12}| = 1474.3$ kJ/kg



Process $1-2$ P = constant

$|W_{23}| = P_2(V_3 - V_2) = 1000(4.152 - 3.741) = 411$ kJ

$$Q_{23} = C_p(T_3 - T_2) = \frac{1.667 \times 2.0771}{(1.667 - 1)}\left(\sqrt{873.15} - 986.60\right) = 449.3 \text{ kJ/kg}$$

$P_2V_2 = RT_2$; $P_3V_3 = RT_3$; $\frac{V_3}{V_2} = \frac{T_3}{T_2}$;

$$V_3 = \frac{873.15}{786.60} \times 3.741 \text{m}^3/\text{kg} = 4.152\ \text{m}^3/\text{kg}$$

Note that $P_1V_1 = RT_1$; $P_1 = \frac{RT_1}{V_1} = \frac{2.0771 \times 313.15}{6.5} = 100.07$ kPa

$$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma} = 313.15\left(\frac{1000}{100.7}\right)^{1.667-1/1.667} = 786.60\ \text{K}$$

$T_1V_1^\gamma = T_2V_2^\gamma$

$$V_2^\gamma = \frac{T_1V_1^\gamma}{T_2} = \left[\frac{T_1V_1^\gamma}{T_2}\right]^{\frac{1}{\gamma}} = \left[\frac{313.15 \times 6.5^{1.667}}{786.60}\right]^{\frac{1}{1.667}} = 3.741\ \text{m}^3/\text{kg}$$

$$m = \frac{P_1V_1}{RT_1} = \frac{100.07 \times 6.5}{2.0771 \times 313.15} = 1.0\ \text{kg}$$

2. Heat engine, $T_H = 973.15$, $T_L = 283.15$

$$\eta_{max} = 1 - \frac{283.15}{973.15} = 70.9\ \%$$

Actual operation $|W| = 200$ kJ $|Q_L| = 100$ kJ

$|Q_H| = |W| + |Q_L| = 300$ kJ

$$\eta_{actual} = \frac{|W|}{Q_H} = \frac{200}{300} = 67\ \%$$

Since $\eta_{actual} < \eta_{optimum}$ the operation is possible

TYPICAL UNIT 3 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS

1. Thermal efficiency of heat engine $=$ network $\frac{\text{output}}{\text{heat}}$ input

$\eta_{TH} = \frac{30}{80} \times 100\ \% = 0.375 \times 100\ \% = \mathbf{37.5}\ \%$

2. $COP_R = \frac{\dot{Q}_L}{\dot{W}_{in}} = \frac{(360/60)}{2} = \mathbf{3}$

3. $COP_{hp} = \frac{\dot{Q}_L}{\dot{W}_{in}}$; $\quad \dot{W}_{in} = \frac{Q}{COP} = \frac{80000\ \text{kJ/h}}{2.5} = \mathbf{32,000\ kJ/h}$

4. $Q = mC_p\Delta T = 1 \times 1.005(35 - 15) = 20.1\ \text{kW}$

$COP_R = \frac{T_L}{T_H - T_L} = \frac{288.15}{308.15 - 288.15} = 14.4075$

Power needed $W = \frac{20.1}{14.4075} = \mathbf{1.40\ kW}$

5. $COP_{hp} = \frac{T_H}{T_H - T_L} = \frac{293.15}{286.84} = 1.022$

Power required $= \dot{W}_{in} = \frac{Q}{COP} = \frac{100}{1.022} = \mathbf{97.85\ kW}$

6. $\dot{W}_{net} = \dot{Q}_H - \dot{Q}_L = 80 - 50 = \mathbf{30\ mW}$

7. $\dot{Q}_{in} = \frac{\dot{W}_{net}}{\eta_{TH}} = \frac{50}{0.24} = 208.3\ \text{kW}$

Fuel consumption rate, $\dot{m}_f = \frac{\dot{Q}_{in}}{CV} = \frac{208.3\ \text{kJ/s}}{44000\ \text{kJ/kg}} = \mathbf{0.00473\ kg/s}$

8. $\dot{Q}_L = 360\ \text{kJ/min}$

$\dot{W}_{in} = 2\frac{\text{kJ}}{\text{s}} = \frac{2\ \text{kJ}}{\left(\frac{1}{60}\right)\text{min}} = 120\ \text{kJ/min}$

$\dot{Q}_H = \dot{W}_{in} + \dot{Q}_L = 360 + 120 = \mathbf{480\ kJ/min}$

9. $\dot{Q}_H = 80{,}000\ \text{kJ/h}$

$COP = 2.5; \quad \dot{W}_{in} = \frac{80000}{2.5} = 32{,}000\ \text{kJ/h}$

$\dot{Q}_L = \dot{Q}_H - \dot{W}_{in} = 80{,}000 - 32{,}000 = \mathbf{48,000\ kJ/h}$

10. $\eta_{Carnot} = 1 - \frac{400}{1000} = 60\ \%$

Actual efficiency $= \frac{300}{325} = 0.615 = 61.5\ \%$

$\eta_{actual} > \eta_{ideal}$, therefore, **this is impossible**